



△ VOLCANIC REMNANT

This oddly shaped island, protruding 165 ft (50 m) above the lake surface, is known as Phantom Ship. Made primarily from a rock called andesite, it is a remnant of a volcano that existed long before Mazama.



Crater Lake

The collapsed crater of an ancient volcano in the Cascade Range now filled by a deep lake

Crater Lake, in south central Oregon, is one of the world's finest examples of a caldera volcano: a stratovolcano that has partially collapsed (usually during the final phase of a catastrophic eruption), creating a gigantic, cauldronlike depression called a caldera. A caldera is sometimes described as a large crater, but it is actually a type of sinkhole, since it forms as a result of collapse rather than explosion. Over time, a caldera often fills with water, resulting in a deep lake like Crater Lake.

Collapse of a stratovolcano

The ancient stratovolcano from which Crater Lake caldera formed is called Mount Mazama. Before its top part collapsed, Mazama stood at around 12,000 ft (3,700 m), making it one of the highest peaks in the Cascade Range. During its eruption (see panel, right), a column of pumice and ash reached some 30 miles (50 km) high. In the final eruption phase, a circular system of cracks, called a ring fracture, opened around the volcano's lower slopes, causing its top to disintegrate.

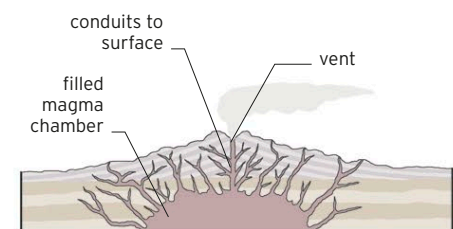
Following this event, volcanic activity continued for some time, creating relatively small volcanic features on the caldera floor. Sediments and material from landslides covered the floor, and the caldera gradually filled with water. Although no significant volcanic activity has been seen at Crater Lake for thousands of years, it is possible that in the future it may erupt again.

◁ PINNACLES

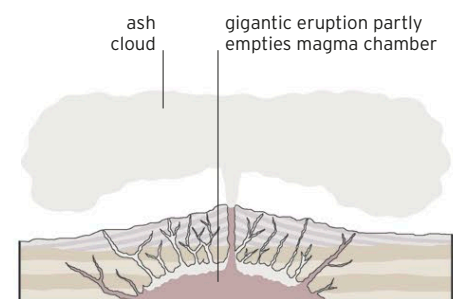
These needlelike formations, a little distance from the lake, were once gas vents under a mass of hot volcanic pumice. The vents hardened into tubes and now stand alone because surrounding softer materials have been eroded away.

HOW CRATER LAKE FORMED

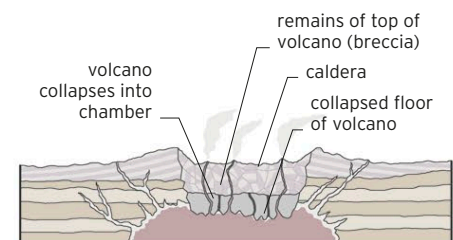
Around 7,700 years ago, a large stratovolcano called Mount Mazama underwent a cataclysmic eruption that left its magma chamber partly empty. Much of the volcano's floor and its top part collapsed downward. This left a huge, bowl-like depression—a caldera—with a raised rim. The caldera later filled with water, forming Crater Lake.



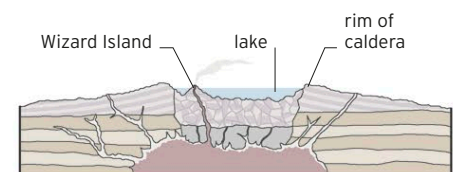
MOUNT MAZAMA 7,700 YEARS AGO



MAGMA CHAMBER PARTLY EMPTIES



VOLCANO COLLAPSES TO FORM CALDERA



A FEW HUNDRED YEARS LATER

At 1,949 ft (594 m) deep, Crater Lake is the **deepest lake** in the **United States**